

Day 13: Area of a Triangle

LESSON + PRACTICE

MathCraft · Grade 6 Mathematics · Professor Xavier Honablué, M.Ed.

Standard: 6.G.A.1 — find the area of right, acute, and obtuse triangles by composing into rectangles or decomposing into triangles; MP.3, MP.6

I can: find the area of any triangle with $A = \frac{1}{2} \times \text{base} \times \text{height}$, pick the height that is perpendicular to the base I chose, and run the formula backwards to find a missing base or height.

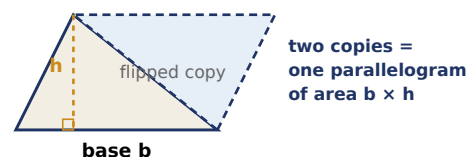
Homework: Earn 90% on IXL "Area of triangles" (GG, 6th grade) — [ixl.com/math/grade-6/area-of-triangles](https://www.ixl.com/math/grade-6/area-of-triangles)

Remember: show ALL work by hand in your math notebook (graph paper is best) and label every answer with units.

Learn

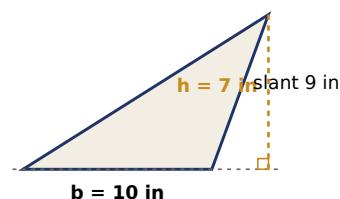
1 · Two triangles make a parallelogram

Take any triangle. Make an exact copy, flip the copy upside down, and slide it against the first one. The two pieces always lock together into a **parallelogram** with the **same base** and the **same height**. That parallelogram's area is $b \times h$ (Day 8), and our triangle is exactly **half** of it. So for every triangle: $A = \frac{1}{2} \times b \times h$, which you can also read as "multiply the base by the height, then cut it in half."



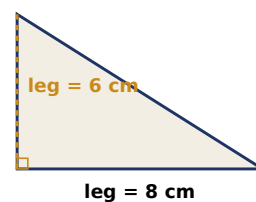
2 · The height is perpendicular — and it may sit outside the triangle

The height is the straight-up distance from the base to the opposite corner, meeting the base at a **right angle**. On a wide (obtuse) triangle, that straight-up line lands **outside** the triangle, so the base is extended with a dotted line. The slanted side is never the height — it is longer, and using it makes the area too big.



3 · In a right triangle, the two legs are the base and the height

A right triangle already has two sides that meet at 90°. Either leg can be the base; the other leg is then the height. The longest side (the one across from the right angle) is never used in the area formula.



4 · Same formula, run backwards

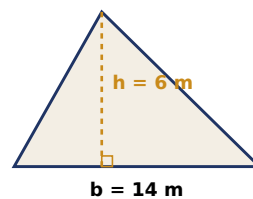
Because $A = \frac{1}{2}bh$, doubling the area undoes the half: $2A = b \times h$. So a missing base is $b = 2A \div h$ and a missing height is $h = 2A \div b$. Always double first — forgetting to double is the most common error on this lesson.

Worked Examples

Example 1. Find the area of a triangle with base 14 m and height 6 m.

- Step 1: $A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 14 \times 6$.
- Step 2: Multiply the base and height first: $14 \times 6 = 84$.
- Step 3: Take half: $84 \div 2 = 42$.
- Step 4: Units: $m \times m = \text{square meters}$.

✓ **Area = 42 sq m**



Example 2. An obtuse triangle has base 10 in, height 7 in, and a slanted side of 9 in. Find the area.

- Step 1: Circle the base (10 in) and the height (7 in). The 9 in slanted side is a distractor.
- Step 2: $A = \frac{1}{2} \times 10 \times 7 = \frac{1}{2} \times 70$.
- Step 3: $70 \div 2 = 35$.
- Step 4: The height landed outside the triangle, but the formula does not change.

✓ **Area = 35 sq in**

Example 3. A triangle has an area of 54 sq ft and a base of 12 ft. Find its height.

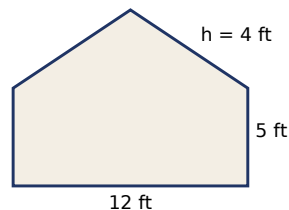
- Step 1: Start from $A = \frac{1}{2} \times b \times h$, so $54 = \frac{1}{2} \times 12 \times h$.
- Step 2: Double both sides to undo the half: $108 = 12 \times h$.
- Step 3: $h = 108 \div 12 = 9$.
- Step 4: Check: $\frac{1}{2} \times 12 \times 9 = 54$. ✓

✓ **Height = 9 ft**

Example 4. A wall is shaped like a rectangle 12 ft wide and 5 ft tall with a triangular top 12 ft wide and 4 ft tall. Find the total area.

- Step 1: Decompose: one rectangle + one triangle.
- Step 2: Rectangle: $12 \times 5 = 60$ sq ft.
- Step 3: Triangle: $\frac{1}{2} \times 12 \times 4 = 24$ sq ft.
- Step 4: Add: $60 + 24 = 84$ sq ft.

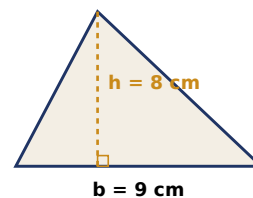
✓ **Area = 84 sq ft**



Practice — show your work in your notebook

1. Find the area of the triangle.

Answer: _____



2. A right triangle has legs of 7 in and 10 in. Find its area.

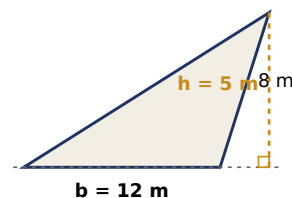
Answer: _____

3. Find the area of a triangle with base 15 ft and height 6 ft.

Answer: _____

4. **Distractor check.** This triangle is labeled base 12 m, height 5 m, and slanted side 8 m. Find its area.

Answer: _____



5. A triangle has an area of 48 sq cm and a height of 8 cm. Find its base.

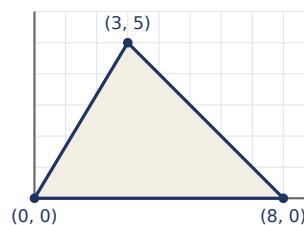
Answer: _____

6. A triangle has an area of 30 sq in and a base of 10 in. Find its height.

Answer: _____

7. Find the area of the triangle plotted on the grid.

Answer: _____



8. **Find the mistake.** For a triangle with base 16 ft and height 9 ft, Andre wrote $A = 16 \times 9 = 144$ sq ft. What did he forget? What is the correct area?

Answer: _____

9. A doghouse wall is a rectangle 10 ft by 6 ft with a triangular roof 10 ft wide and 4 ft tall. Find the total area.

Answer: _____

10. A triangular garden has a base of 18 ft and a height of 7 ft. One bag of grass seed covers 30 sq ft. How many whole bags are needed?

Answer: _____

11. Find the area of a triangle with base 7.5 cm and height 4 cm.

Answer: _____

12. Find the area of a triangle with base 5 in and height $3\frac{1}{2}$ in.

Answer: _____

Answer Key & Worked Solutions

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1. Find the area of the triangle.

- Step 1: $A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 9 \times 8$.
- Step 2: $9 \times 8 = 72$.
- Step 3: $72 \div 2 = 36$.

✓ **Answer: 36 sq cm**

2. A right triangle has legs of 7 in and 10 in. Find its area.

- Step 1: In a right triangle the two legs are the base and the height.
- Step 2: $A = \frac{1}{2} \times 7 \times 10 = \frac{1}{2} \times 70 = 35$.

✓ **Answer: 35 sq in**

3. Find the area of a triangle with base 15 ft and height 6 ft.

- Step 1: $A = \frac{1}{2} \times 15 \times 6$.
- Step 2: $15 \times 6 = 90$.
- Step 3: $90 \div 2 = 45$.

✓ **Answer: 45 sq ft**

4. Distractor check. This triangle is labeled base 12 m, height 5 m, and slanted side 8 m. Find its area.

- Step 1: Use only the base and the perpendicular height: 12 and 5.
- Step 2: $A = \frac{1}{2} \times 12 \times 5 = \frac{1}{2} \times 60 = 30$.
- Step 3: The 8 m slanted side is not used ($12 \times 8 \div 2 = 48$ is the trap answer).

✓ **Answer: 30 sq m**

5. A triangle has an area of 48 sq cm and a height of 8 cm. Find its base.

- Step 1: $48 = \frac{1}{2} \times b \times 8$.
- Step 2: Double the area first: $96 = b \times 8$.
- Step 3: $b = 96 \div 8 = 12$.

✓ **Answer: 12 cm**

6. A triangle has an area of 30 sq in and a base of 10 in. Find its height.

- Step 1: Double the area: $60 = 10 \times h$.
- Step 2: $h = 60 \div 10 = 6$.

✓ **Answer: 6 in**

7. Find the area of the triangle plotted on the grid.

- Step 1: Base runs from (0, 0) to (8, 0): 8 units.
- Step 2: Height is the straight-up distance to $y = 5$: 5 units.
- Step 3: $A = \frac{1}{2} \times 8 \times 5 = 20$.

✓ **Answer: 20 square units**

8. Find the mistake. For a triangle with base 16 ft and height 9 ft, Andre wrote $A = 16 \times 9 = 144$ sq ft. What did he forget? What is the correct area?

- Step 1: Andre found the area of the whole parallelogram, not the triangle.
- Step 2: A triangle is half of it: $144 \div 2 = 72$.
- Step 3: He forgot the $\frac{1}{2}$.

✓ **Answer: He left out the $\frac{1}{2}$; correct area is 72 sq ft**

9. A doghouse wall is a rectangle 10 ft by 6 ft with a triangular roof 10 ft wide and 4 ft tall. Find the total area.

- Step 1: Rectangle: $10 \times 6 = 60$.
- Step 2: Triangle: $\frac{1}{2} \times 10 \times 4 = 20$.
- Step 3: $60 + 20 = 80$.

✓ **Answer: 80 sq ft**

10. A triangular garden has a base of 18 ft and a height of 7 ft. One bag of grass seed covers 30 sq ft. How many whole bags are needed?

- Step 1: Area: $\frac{1}{2} \times 18 \times 7 = \frac{1}{2} \times 126 = 63$ sq ft.
- Step 2: $63 \div 30 = 2.1$ bags.
- Step 3: You cannot buy part of a bag, so round up to 3.

✓ **Answer: 3 bags (63 sq ft to cover)**

11. Find the area of a triangle with base 7.5 cm and height 4 cm.

- Step 1: $A = \frac{1}{2} \times 7.5 \times 4$.
- Step 2: $7.5 \times 4 = 30$ ($75 \times 4 = 300$, one decimal place).
- Step 3: $30 \div 2 = 15$.

✓ **Answer: 15 sq cm**

12. Find the area of a triangle with base 5 in and height $3\frac{1}{2}$ in.

- Step 1: $A = \frac{1}{2} \times 5 \times 3\frac{1}{2}$.
- Step 2: $5 \times 3\frac{1}{2} = 5 \times 3 + 5 \times \frac{1}{2} = 15 + 2\frac{1}{2} = 17\frac{1}{2}$.
- Step 3: Half of $17\frac{1}{2} = 8\frac{3}{4}$.

✓ **Answer: $8\frac{3}{4}$ sq in**